Redwood Regional Bank CECL Review Memo

March 13, 2026

Executive Conclusion

This memo documents an end-to-end CECL review of Redwood Regional Bank's Q1 2026 CECL Allowance Review. The supplied reserve engine was executable, the packaged baseline reserve was reproduced exactly, and scenario reruns plus targeted sensitivities were completed using the supplied portfolio, scenario, and overlay artifacts. Portfolio-level reserve direction was broadly reasonable with total ACL rising from $4,876,841 under Baseline to $7,132,022 under Adverse and $8,287,703 under Severe. The review nonetheless identified material exceptions requiring remediation before an unqualified approval posture would be appropriate.

- Review status: qualified internal review conclusion; package supports substantive CECL challenge work, but identified findings should be remediated before an unqualified governance posture is taken.
- Most material quantitative observation: portfolio reserve ordering is monotonic, but Residential Mortgage reserve declines from Adverse to Severe.
- Most material documentation observation: documented forecast and reversion treatment do not match the implemented reserve engine.
- Most material governance observation: overlay usage is materially larger than the documented cap.

Review Scope, Package Reviewed, and Coverage

This memo documents an internal CECL review of the supplied package for Redwood Regional Bank's Q1 2026 CECL Allowance Review. The work focused on whether the package was internally coherent, reproducible, directionally reasonable under stressed scenarios, and aligned to its own methodology and governance materials.

- Supported work: baseline reproduction, portfolio and segment scenario reruns, sensitivity testing, reserve-driver interpretation, and documentation/governance alignment review.
- Blocked work: none within the scope of the supplied package.
- Coverage boundary: conclusions are bounded to the uploaded CECL package, the synthetic portfolio, and the scenarios and overlays included in that package.

- Constraint: No independent historical performance dataset or external benchmark was supplied; conclusions are bounded to the package and synthetic portfolio.

Materials Reviewed

The uploaded package contained enough material to support both quantitative reruns and documentation challenge. The artifact groups below reflect the main categories of bank-supplied inputs that informed the review.

Model implementation

- `model/cecl_engine.py`

Portfolio data and overlays

- `data/data_dictionary.csv`
- `data/loan_level_snapshot.csv`
- `data/overlay_schedule.csv`

Scenario definitions

- `scenarios/adverse.csv`
- `scenarios/baseline.csv`
- `scenarios/severe.csv`

Documentation and governance materials

- `docs/control_process_note.md`
- `docs/governance_minutes.md`
- `docs/methodology.md`
- `docs/model_overview.md`
- `docs/overlay_memo.md`
- `docs/prior_review_note.md`
- `docs/scenario_assumptions.md`

Packaged prior outputs

- `outputs/prior_baseline_reserve.csv`
- `outputs/prior_scenario_summary.csv`

- `outputs/prior_segment_reserves.csv`

CECL Process Overview

The discovered package is best understood as a model-driven CECL review case for Redwood Regional Bank's Q1 2026 CECL Allowance Review. It includes executable reserve logic, loan-level portfolio data, scenario definitions, prior baseline outputs, and enough supporting documentation to support both quantitative reruns and documentation challenge. The documented CECL process covers 4 primary segments: Commercial Real Estate, Commercial and Industrial, Residential Mortgage, Consumer Unsecured. The reserve process applies scenario-dependent macro paths to loan-level reserve components and then layers scenario-specific qualitative overlays by segment.

Assumption Area	Documented Position	Observed in Package	Review Implication
Forecast horizon	4 quarters documented	6 quarters implemented	Methodology and model overview are not aligned to engine behavior.
Reversion horizon	4 quarters documented	2 quarters implemented	Policy description and actual reserve mechanics differ.
Overlay cap	5.0 bps documented	9.0 bps scheduled	Qualitative adjustment magnitude exceeds documented guardrail.
Scenario reasonableness	Severe should be directionally harsher unless clearly justified	Residential Mortgage reserve falls by $109,988 from adverse to severe	Segment-level severe path requires additional governance challenge.

Validation Approach

Codex formed the review strategy from the discovered evidence. The package supported 10 executable procedures across quantitative and documentation lanes, with 0 blocked procedures. Because the package contained runnable code, scenario files, prior outputs, and core methodology documents, the review strategy prioritized execution-based challenge first and then used documentation review to test whether the written methodology and governance artifacts actually described the observed reserve behavior.

1. Does the package contain enough evidence to support a full execution-based CECL review rather than a documentation-only assessment?
2. Can the supplied baseline reserve be reproduced from the packaged reserve engine, portfolio data, and baseline scenario?
3. Do reserve outputs move directionally and materially across Baseline, Adverse, and Severe scenarios both overall and by major segment?
4. How sensitive are reserves to forecast horizon, reversion speed, macro severity, and overlay magnitude assumptions?
5. Do the methodology, scenario, and overlay documents faithfully describe the behavior observed in the implemented CECL process, especially for Residential Mortgage?

ID	Procedure	What Codex Was Trying To Test	Status
FR-01	Package inventory and case classification	Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.	Executed
FR-02	Baseline reproduction	Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.	Executed
FR-03	Portfolio-level scenario reruns	Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.	Executed
FR-04	Segment-level reasonableness review	Test whether scenario ordering remains directionally sensible at the segment level.	Executed
FR-05	Forecast horizon sensitivity	Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon.	Executed
FR-06	Reversion speed sensitivity	Assess how severe-scenario reserve responds to longer reversion periods.	Executed
FR-07	Macro severity sensitivity	Test whether reserve responds proportionally to macro severity scaling.	Executed
FR-08	Overlay magnitude sensitivity	Assess the reserve effect of removing or amplifying qualitative overlays.	Executed
FR-09	Reserve driver bridge	Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.	Executed
FR-10	Methodology and documentation alignment review	Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.	Executed

Detailed planning rationale, evidence mapping, and the granular procedure log are retained in the appendices and support files rather than in the main body of the memo.

Quantitative Review Results

Baseline Reproduction

Baseline reproduction landed at $4,876,841 against a packaged baseline of $4,876,841, with absolute delta $0 and maximum loan-level delta $0. Relative to Baseline, Adverse increased reserve by $2,255,180 and Severe increased reserve by another $1,155,682, for a total Severe-versus-Baseline increase of $3,410,862.

Metric	Result
Packaged baseline reserve	$4,876,841
Rerun baseline reserve	$4,876,841
Absolute delta	$0
Percent delta	0.000%
Maximum loan-level absolute delta	$0

Scenario Analysis

Portfolio reserve ordering was monotonic. Adverse increased reserve by $2,255,180 versus Baseline, and Severe increased reserve by $1,155,682 versus Adverse.

Scenario	Reserve	Reserve Rate	Change vs Baseline
Baseline	$4,876,841	0.60%	$0
Adverse	$7,132,022	0.88%	$2,255,180
Severe	$8,287,703	1.03%	$3,410,862

Segment-Level Reasonableness

Most segments behaved directionally as expected under worsening macro conditions. The principal exception was Residential Mortgage, which produced a severe-scenario reserve below its adverse reserve.

Segment	Baseline	Adverse	Severe	Sev - Adv
Commercial And Industrial	$1,977,246	$2,712,809	$3,148,324	$435,515
Commercial Real Estate	$2,545,520	$3,972,723	$4,787,770	$815,047
Consumer Unsecured	$100,356	$118,791	$133,899	$15,108
Residential Mortgage	$253,720	$327,699	$217,711	$-109,988

Sensitivity Analysis

Sensitivity testing showed limited change from moving the documented forecast horizon to 4 quarters (change of $1,145), but reserve meaningfully increased as reversion lengthened from 2 to 6 quarters under Severe (change of $185,201). Scaling severe macro stress from 1.00x to 1.25x added $1,515,402, while removing overlays reduced Severe reserve by $436,787.

- Forecast horizon sensitivity: moving from 6 to 4 forecast quarters changed reserve by $1,145.
- Reversion sensitivity: extending severe-scenario reversion from 2 to 6 quarters increased reserve by $185,201.
- Macro severity sensitivity: scaling severe stress from 1.00x to 1.25x increased reserve by $1,515,402.
- Overlay sensitivity: removing overlays reduced reserve by $436,787.

The full sensitivity grid is retained in the support files and should be used for any follow-on appendix work or management discussion.

Reserve Driver Interpretation

The reserve-driver bridge indicates that Cre Price Growth was the largest modeled contributor to the Severe-versus-Baseline reserve increase, adding $1,108,566 on a standalone basis.

Driver	Reserve Delta vs Baseline
Cre Price Growth	$1,108,566
Gdp Growth	$455,249
Overlay	$436,787
Unemployment Rate	$336,020

Quantitative Workpapers

- `outputs/support/baseline_reproduction.json`
- `outputs/support/scenario_results.csv`
- `outputs/support/segment_reserve_comparison.csv`
- `outputs/support/sensitivity_results.csv`
- `outputs/support/driver_bridge.csv`

Documentation and Governance Review

Documentation cross-checking found that the written methodology still describes a 4-quarter reasonable-and-supportable period with 4-quarter reversion, while the engine implements 6 forecast quarters and 2 reversion quarters. The overlay memo references a 5.0 bps guardrail, but the supplied overlay schedule applies up to 9.0 bps. At segment level, Residential Mortgage shows a severe-scenario reserve $217,711, below its adverse reserve $327,699, driven by the scenario and overlay posture described in the package.

- Documented forecast/reversion: 4Q forecast and 4Q reversion.
- Implemented forecast/reversion: 6Q forecast and 2Q reversion.
- Documented overlay cap: 5.0 bps; scheduled overlay cap: 9.0 bps.
- Scenario/segment observation: Residential Mortgage severe reserve $217,711 versus adverse reserve $327,699.

Documentation and Governance Workpapers

- `outputs/support/documentation_crosscheck.md`
- `outputs/support/findings_register.json`
- `outputs/support/evidence_excerpts.md`

Findings and Required Remediation

Documented CECL horizon and reversion do not match the implemented reserve engine

Severity: HIGH
Observation: Documentation describes a 4-quarter reasonable-and-supportable period with 4-quarter reversion, while the reserve engine is configured for 6 forecast quarters and 2 reversion quarters.
Evidence: docs/methodology.md, docs/model_overview.md, model/cecl_engine.py
Recommended response: Either update the methodology and model overview to the implemented 6-quarter forecast and 2-quarter reversion, or change the engine to match documented policy before the next governance submission.

Qualitative overlay effect is materially larger than documentation suggests

Severity: MEDIUM
Observation: The documentation frames qualitative overlays as capped at 5.0 bps, but the supplied overlay table uses up to 9.0 bps.
Evidence: docs/overlay_memo.md, data/overlay_schedule.csv, outputs/support/sensitivity_results.csv
Recommended response: Reconcile the overlay memo and governance materials to the actual schedule, including the 9.0 bps cap and the rationale for segment-level relief under Severe.

One segment behaves directionally oddly under the severe scenario

Severity: HIGH
Observation: Segment residential_mortgage produces a severe-scenario reserve of 217710.91, below the adverse reserve of 327699.21. The severe path allows house prices to recover faster than adverse after 2026Q4 while the overlay schedule applies negative mortgage relief, creating a directionally odd segment result.
Evidence: outputs/support/segment_reserve_comparison.csv, scenarios/severe.csv, docs/scenario_assumptions.
docs/overlay_memo.md
Recommended response: Challenge the Residential Mortgage severe scenario result, document whether the house-price recovery and negative overlay are intended, and update committee materials accordingly.

Baseline reserve was reproducible within tolerance

Severity: LOW
Observation: Rerunning the supplied baseline logic reproduced the packaged baseline reserve within 0.000%.
Evidence: outputs/prior_baseline_reserve.csv, outputs/support/baseline_reproduction.json
Recommended response: Retain the reproduction evidence as support for package completeness and execution readiness.

Overall Assessment

The supplied package supported a substantive CECL review and allowed execution of the core quantitative and documentation work expected for this scope. The baseline output was reproducible and overall reserve direction was reasonable, but documentation and governance artifacts are not yet aligned to implemented behavior, and one segment-level severe result requires explicit management challenge before the package would support an unqualified review conclusion.

Appendix A. Materials Reviewed and Artifact Provenance

The main body summarizes the material inputs and outputs. This appendix records the provenance split more explicitly so a reviewer can distinguish what the bank uploaded from what Codex generated during the review.

Bank-Uploaded Input Categories

Model implementation

- `model/cecl_engine.py`

Portfolio data and overlays

- `data/data_dictionary.csv`
- `data/loan_level_snapshot.csv`
- `data/overlay_schedule.csv`

Scenario definitions

- `scenarios/adverse.csv`
- `scenarios/baseline.csv`
- `scenarios/severe.csv`

Documentation and governance materials

- `docs/control_process_note.md`
- `docs/governance_minutes.md`
- `docs/methodology.md`
- `docs/model_overview.md`
- `docs/overlay_memo.md`

- docs/prior_review_note.md
- docs/scenario_assumptions.md

Packaged prior outputs

- outputs/prior_baseline_reserve.csv
- outputs/prior_scenario_summary.csv
- outputs/prior_segment_reserves.csv

Codex-Generated Review Categories

Discovery and planning workpapers

- outputs/support/discovery_summary.json
- outputs/support/discovery_summary.md
- outputs/support/case_understanding.md
- outputs/support/review_plan.md
- outputs/support/review_strategy.md
- outputs/support/executed_test_matrix.md
- outputs/support/executed_test_matrix.csv
- outputs/support/evidence_map.md
- outputs/support/evidence_ledger.json
- outputs/support/coverage_statement.md

Quantitative workpapers

- outputs/support/baseline_reproduction.json
- outputs/support/scenario_results.csv
- outputs/support/segment_reserve_comparison.csv
- outputs/support/sensitivity_results.csv
- outputs/support/driver_bridge.csv

Documentation and findings workpapers

- outputs/support/evidence_excerpts.md
- outputs/support/documentation_crosscheck.md
- outputs/support/findings_register.json

Process and trace artifacts

- `outputs/support/agentic_review_log.md`
- `outputs/support/procedure_run_log.md`
- `outputs/support/procedure_run_log.csv`
- `outputs/support/codex_trace.md`

Full Provenance Registers

- `outputs/support/artifact_provenance.md`
- `outputs/support/artifact_provenance.json`
- `outputs/support/evidence_ledger.json`

Appendix B. Executed Procedure Register

FR-01 - Package inventory and case classification

Determine whether the upload supports a model-driven CECL review and identify the core evidence types available.

- **Status:** executed
- **Why selected:** Runnable code, scenario tables, loan-level data, prior outputs, and methodology documents were all present.
- **Bank-input evidence relied upon:** `model/cecl_engine.py`, `data/loan_level_snapshot.csv`, `scenarios/baseline.csv`, `outputs/prior_baseline_reserve.csv`, `docs/methodology.md`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `discovery_summary.md`, `evidence_ledger.json`
- **Assessment criteria:** Review should only proceed as execution-based if engine, data, scenarios, and prior outputs are all available.
- **Key result:** Package classified as model-driven CECL review with no material execution blockers.

FR-02 - Baseline reproduction

Confirm that the packaged baseline reserve can be reproduced from the supplied engine and portfolio inputs.

- **Status:** executed
- **Why selected:** A packaged prior baseline reserve was supplied alongside the runnable engine and source portfolio.

- **Bank-input evidence relied upon:** `model/cecl_engine.py`, `data/loan_level_snapshot.csv`, `outputs/prior_baseline_reserve.csv`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `baseline_reproduction.json`
- **Assessment criteria:** Total reserve and segment reserve should reconcile within tolerance; large deltas would undermine confidence in package integrity.
- **Key result:** Baseline rerun matched packaged reserve exactly at 4876841.47 with 0.000% delta.

FR-03 - Portfolio-level scenario reruns

Run Baseline, Adverse, and Severe scenarios and assess directional reserve ordering.

- **Status:** executed
- **Why selected:** The package included three explicit scenario files and a model designed to consume them.
- **Bank-input evidence relied upon:** `scenarios/baseline.csv`, `scenarios/adverse.csv`, `scenarios/severe.csv`, `model/cecl_engine.py`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `scenario_results.csv`
- **Assessment criteria:** Portfolio reserve should generally increase with scenario severity absent a documented exception.
- **Key result:** Portfolio reserve increased from 4876841.47 to 7132021.97 to 8287703.47.

FR-04 - Segment-level reasonableness review

Test whether scenario ordering remains directionally sensible at the segment level.

- **Status:** executed
- **Why selected:** CECL governance depends on segment-level reserve understanding, not just portfolio totals.
- **Bank-input evidence relied upon:** `docs/scenario_assumptions.md`, `docs/overlay_memo.md`
- **Codex-generated evidence relied upon:** `outputs/support/segment_reserve_comparison.csv`
- **Codex-generated outputs produced:** `segment_reserve_comparison.csv`
- **Assessment criteria:** Segment movements should generally align to scenario severity or be explicitly supported by documented assumptions.
- **Key result:** Residential Mortgage severe reserve was 217710.91, below adverse reserve 327699.21, creating a documented challenge item.

FR-05 - Forecast horizon sensitivity

Assess reserve sensitivity to changing the reasonable-and-supportable forecast horizon.

- **Status:** executed
- **Why selected:** The methodology makes forecast horizon a named governance assumption.
- **Bank-input evidence relied upon:** `docs/methodology.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Reserve movement should be understandable and documented; mismatch to stated policy must be surfaced.
- **Key result:** Moving from 6 to 4 forecast quarters changed baseline reserve only modestly, but it directly exposed a documented-versus-implemented policy mismatch.

FR-06 - Reversion speed sensitivity

Assess how severe-scenario reserve responds to longer reversion periods.

- **Status:** executed
- **Why selected:** Reversion mechanics are central to CECL governance and are explicitly documented in the package.
- **Bank-input evidence relied upon:** `docs/methodology.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Reserve should move directionally as reversion assumptions are extended or shortened.
- **Key result:** Extending severe-scenario reversion from 2 to 6 quarters increased reserve materially, confirming reversion is a meaningful reserve lever.

FR-07 - Macro severity sensitivity

Test whether reserve responds proportionally to macro severity scaling.

- **Status:** executed
- **Why selected:** Scenario severity and directional reasonableness are core to CECL challenge work.
- **Bank-input evidence relied upon:** `scenarios/severe.csv`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`

- **Assessment criteria:** Scaling macro severity upward should generally increase reserve; weaker response would require explanation.
- **Key result:** Scaling severe stress from 1.00x to 1.25x materially increased reserve, supporting macro sensitivity monotonicity at portfolio level.

FR-08 - Overlay magnitude sensitivity

Assess the reserve effect of removing or amplifying qualitative overlays.

- **Status:** executed
- **Why selected:** The package included a scenario-by-segment overlay schedule and an overlay governance memo.
- **Bank-input evidence relied upon:** `data/overlay_schedule.csv`, `docs/overlay_memo.md`
- **Codex-generated evidence relied upon:** `outputs/support/sensitivity_results.csv`
- **Codex-generated outputs produced:** `sensitivity_results.csv`
- **Assessment criteria:** Overlay effect should be quantitatively explainable and consistent with governance statements.
- **Key result:** Removing severe overlays reduced reserve materially, confirming overlay posture is consequential and governance-relevant.

FR-09 - Reserve driver bridge

Identify which macro and overlay drivers explain the majority of severe-versus-baseline reserve movement.

- **Status:** executed
- **Why selected:** A driver bridge makes the review more interpretable and supports reasonableness challenge.
- **Bank-input evidence relied upon:** `scenarios/baseline.csv`, `scenarios/severe.csv`
- **Codex-generated evidence relied upon:** `outputs/support/driver_bridge.csv`
- **Codex-generated outputs produced:** `driver_bridge.csv`
- **Assessment criteria:** Major reserve movement should be attributable to understandable drivers rather than unexplained residual behavior.
- **Key result:** CRE price growth was the largest modeled standalone driver of severe-versus-baseline reserve change.

FR-10 - Methodology and documentation alignment review

Test whether written methodology, model overview, and overlay documentation match implemented behavior and observed outputs.

- **Status:** executed
- **Why selected:** The package contained enough documentation to support direct cross-checking against the model and results.
- **Bank-input evidence relied upon:** `docs/methodology.md`, `docs/model_overview.md`, `docs/scenario_ass` `docs/overlay_memo.md`, `model/cecl_engine.py`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `documentation_crosscheck.md`, `findings_register.json`
- **Assessment criteria:** Documented horizon, reversion, scenario logic, and overlay posture should match the implemented process or be clearly qualified.
- **Key result:** Cross-checking identified a forecast/reversion mismatch, overlay cap mismatch, and a severe-scenario segment anomaly requiring remediation.

Appendix C. Agentic Review Record

This appendix records the staged review behavior and the granular action log behind the memo. It is intended to show the review workflow without interrupting the main validation narrative.

Discovery

The review started by inventorying the package and determining whether a quantitative review was actually possible from the supplied materials.

- Review question addressed: What is in the package and does it support execution-based CECL review?
- Decision or rationale: Treat the case as model-driven because reserve code, loan-level data, scenarios, prior outputs, and core methodology documents were all present.
- Inputs reviewed: input_package/*
- Outputs produced: discovery_summary.md, evidence_ledger.json, case_understanding.md

Case understanding and planning

Codex formed a review strategy around the strongest available evidence and explicitly selected a test set before running the work.

- Review question addressed: Which review questions matter most for this package and which procedures can answer them credibly?
- Decision or rationale: Prioritize baseline reproduction, scenario reruns, sensitivities, and documentation alignment because the package supports all four lanes.
- Inputs reviewed: discovery_summary.md, docs/methodology.md, docs/scenario_assumptions.md, docs/overlay_memo.md

- Outputs produced: review_plan.md, review_strategy.md, executed_test_matrix.md

Quantitative execution

The review ran the CECL engine against baseline and stressed scenarios, then executed targeted sensitivities to challenge the major reserve assumptions.

- Review question addressed: Can the reserve process be reproduced, and do outputs move sensibly overall and by segment?
- Decision or rationale: Run baseline, adverse, and severe scenarios first, then test horizon, reversion, macro severity, and overlays as the most consequential levers.
- Inputs reviewed: model/cecl_engine.py, data/loan_level_snapshot.csv, scenarios/*.csv, data/overlay_schedule
- Outputs produced: baseline_reproduction.json, scenario_results.csv, segment_reserve_comparison.csv, sensitivity_results.csv, driver_bridge.csv

Documentation challenge

After quantitative outputs were produced, the review cross-checked the documents against implemented behavior and reserve results.

- Review question addressed: Do the written methodology, scenario, and overlay materials describe the same process that the results imply?
- Decision or rationale: Escalate mismatches where documented governance assumptions differ from implementation or where narrative support is insufficient for observed output behavior.
- Inputs reviewed: docs/*.md, model/cecl_engine.py, segment_reserve_comparison.csv, sensitivity_results.csv
- Outputs produced: documentation_crosscheck.md, findings_register.json, evidence_map.md

Synthesis

The final stage combined discovery, planning, quantitative results, and documentation challenge into an internal review report and support pack.

- Review question addressed: What opinion can be supported, what findings should be issued, and what evidence supports that conclusion?
- Decision or rationale: Issue a substantive review with explicit remediation for documentation mismatch, overlay governance, and the residential mortgage severe-scenario anomaly.
- Inputs reviewed: findings_register.json, executed_test_matrix.md, coverage_statement.md, agentic_review_log.md
- Outputs produced: cecl_review_memo.tex, cecl_review_memo.pdf

Planning and Process Files

- `outputs/support/artifact_provenance.md`
- `outputs/support/review_plan.md`
- `outputs/support/review_strategy.md`
- `outputs/support/executed_test_matrix.md`
- `outputs/support/procedure_run_log.md`
- `outputs/support/codex_trace.md`

Chronological Procedure Log

EVT-01 - Inventory uploaded package

- **Phase:** discovery
- **Related procedure:** FR-01
- **Action:** Enumerated the input package and confirmed presence of reserve engine, loan-level data, scenarios, prior outputs, and CECL methodology documents.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/*.csv`, `input_package/docs/*.md`
- **Outputs:** `outputs/support/discovery_summary.md`, `outputs/support/evidence_ledger.json`
- **Result:** Case classified as execution-capable CECL review with no material runtime blockers.

EVT-02 - Select review lanes

- **Phase:** planning
- **Related procedure:** FR-01
- **Action:** Chose baseline reproduction, scenario reruns, sensitivity testing, reserve bridge, and documentation challenge as the primary review lanes.
- **Inputs:** `outputs/support/discovery_summary.md`, `input_package/docs/methodology.md`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/review_plan.md`, `outputs/support/review_strategy.md`
- **Result:** Ten executable procedures selected; none blocked.

EVT-03 - Reproduce packaged baseline reserve

- **Phase:** execution
- **Related procedure:** FR-02

- **Action:** Reran the packaged engine under the baseline scenario and reconciled the reproduced reserve to the supplied baseline reserve file.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/baseline.csv`, `input_package/outputs/prior_baseline_reserve.csv`
- **Outputs:** `outputs/support/baseline_reproduction.json`
- **Result:** Packaged reserve 4876841.47; rerun reserve 4876841.47; absolute delta 0.00.

EVT-04 - Run baseline, adverse, and severe scenarios

- **Phase:** execution
- **Related procedure:** FR-03
- **Action:** Executed the reserve engine across the three supplied scenario paths and aggregated total reserve results.
- **Inputs:** `input_package/model/cecl_engine.py`, `input_package/data/loan_level_snapshot.csv`, `input_package/scenarios/baseline.csv`, `input_package/scenarios/adverse.csv`, `input_package/scena`
- **Outputs:** `outputs/support/scenario_results.csv`
- **Result:** Portfolio reserve moved from 4876841.47 to 7132021.97 to 8287703.47.

EVT-05 - Check segment-level scenario ordering

- **Phase:** execution
- **Related procedure:** FR-04
- **Action:** Compared segment reserve amounts across baseline, adverse, and severe scenarios to test directional reasonableness by segment.
- **Inputs:** `outputs/support/scenario_results.csv`, `input_package/docs/scenario_assumptions.md`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/segment_reserve_comparison.csv`
- **Result:** Residential Mortgage reserve fell from 327699.21 in adverse to 217710.91 in severe.

EVT-06 - Run forecast-horizon sensitivity

- **Phase:** execution
- **Related procedure:** FR-05
- **Action:** Tested baseline reserve under alternate forecast horizons to challenge the documented 4-quarter assumption versus the 6-quarter implementation.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`

- **Result:** Changing the baseline forecast horizon from 6 quarters to 4 quarters changed reserve modestly, but confirmed a policy-versus-implementation mismatch.

EVT-07 - Run reversion-speed sensitivity

- **Phase:** execution
- **Related procedure:** FR-06
- **Action:** Re-estimated severe-scenario reserve using longer reversion periods to test sensitivity to the reversion assumption.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Extending severe reversion from 2 to 6 quarters increased reserve from 8287703.47 to 8472904.31.

EVT-08 - Run macro-severity scaling sensitivity

- **Phase:** execution
- **Related procedure:** FR-07
- **Action:** Scaled severe macro stress to test whether reserve responds monotonically to harsher macro conditions.
- **Inputs:** `input_package/scenarios/severe.csv`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Increasing severe macro scaling from 1.00x to 1.25x increased reserve from 8287703.47 to 9803105.55.

EVT-09 - Run overlay-magnitude sensitivity

- **Phase:** execution
- **Related procedure:** FR-08
- **Action:** Removed and amplified overlays to measure how much of the severe reserve outcome is attributable to overlay posture.
- **Inputs:** `input_package/data/overlay_schedule.csv`, `input_package/docs/overlay_memo.md`
- **Outputs:** `outputs/support/sensitivity_results.csv`
- **Result:** Removing severe overlays reduced reserve from 8287703.47 to 7850916.88; amplifying to 1.50x raised reserve to 8506096.88.

EVT-10 - Build reserve-driver bridge

- **Phase:** execution
- **Related procedure:** FR-09
- **Action:** Constructed a simple driver bridge to attribute severe-versus-baseline reserve movement to major macro and overlay factors.
- **Inputs:** `input_package/scenarios/baseline.csv`, `input_package/scenarios/severe.csv`, `input_package/model/cecl_engine.py`
- **Outputs:** `outputs/support/driver_bridge.csv`
- **Result:** CRE price growth was the largest standalone driver, contributing 1108566.01 of severe-versus-baseline reserve change.

EVT-11 - Cross-check methodology against implementation

- **Phase:** documentation_challenge
- **Related procedure:** FR-10
- **Action:** Compared documented forecast, reversion, scenario, and overlay assumptions to the implemented engine constants and quantitative outputs.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/docs/model_overview.md`, `input_package/docs/overlay_memo.md`, `input_package/model/cecl_engine.py`, `outputs/support/segmen`
- **Outputs:** `outputs/support/documentation_crosscheck.md`, `outputs/support/findings_register.json`
- **Result:** Recorded horizon/reversion mismatch, overlay-cap mismatch, and severe Residential Mortgage anomaly as formal findings.

EVT-12 - Assemble internal review pack

- **Phase:** synthesis
- **Related procedure:** FR-10
- **Action:** Combined discovery, planning, quantitative outputs, documentation challenge, and evidence mapping into the final internal-review report pack.
- **Inputs:** `outputs/support/review_strategy.md`, `outputs/support/procedure_run_log.md`, `outputs/support/findings_register.json`, `outputs/support/evidence_map.md`
- **Outputs:** `outputs/stakeholder/cecl_review_memo.tex`, `outputs/stakeholder/cecl_review_memo.pdf`
- **Result:** Produced a comprehensive internal CECL review memo with explicit provenance, raw evidence excerpts, and procedure-level detail.

Appendix D. Selected Evidence Excerpts

Only text and code excerpts are reproduced here. Structured data outputs are referenced through the support files listed elsewhere in the memo.

[BANK INPUT] docs/methodology.md

Documented CECL forecast and reversion assumptions cited in the horizon mismatch finding.

```
# CECL Methodology

## Objective
Review the Q1 2026 CECL Allowance Review CECL reserve process for Redwood Regional Bank. The documented

## Segments
- Commercial Real Estate
- Commercial and Industrial
- Residential Mortgage
- Consumer Unsecured

## Key Drivers
- Unemployment rate
- GDP growth
- House price growth
- CRE price growth
- Prime rate

## Overlay Treatment
Qualitative overlays are documented as targeted adjustments capped at 5.0 bps and are not expected to cl
```

[BANK INPUT] model/cecl_engine.py

Implemented reserve-engine constants and overlay schedule that Codex compared to the methodology and overlay memo.

```
"cre_price_growth": 0.0,
      "prime_rate": 0.11
    }
  }
}
OVERLAYS = {
  "baseline": {
    "commercial_real_estate": 2.0,
    "commercial_and_industrial": 2.0,
    "residential_mortgage": 1.0,
    "consumer_unsecured": 3.0
  },
  "adverse": {
    "commercial_real_estate": 5.0,
    "commercial_and_industrial": 4.0,
```

```
    "residential_mortgage": 2.0,
    "consumer_unsecured": 6.0
  },
  "severe": {
    "commercial_real_estate": 8.0,
    "commercial_and_industrial": 7.0,
    "residential_mortgage": -9.0,
    "consumer_unsecured": 9.0
  }
}
FORECAST_QUARTERS = 6
REVERSION_QUARTERS = 2


def build_projection(frame: pd.DataFrame) -> pd.DataFrame:
    long_run = {
        "unemployment_rate": 4.7,
        "gdp_growth": 1.7,
        "house_price_growth": 1.9,
        "cre_price_growth": 1.4,
        "prime_rate": 5.0,
    }
    forecast = frame.iloc[:FORECAST_QUARTERS].copy()
    last = forecast.iloc[-1].copy()
    reversion_rows = []
    for step in range(1, REVERSION_QUARTERS + 1):
        weight = step / REVERSION_QUARTERS
        row = {"quarter": f"reversion_{step}"}
        for key, long_run_value in long_run.items():
            row[key] = round(float(last[key]) + (long_run_value - float(last[key])) * weight, 4)
        reversion_rows.append(row)
```

[BANK INPUT] docs/overlay_memo.md

Documented overlay cap and severe overlay posture used in the overlay-governance challenge.

```
# Overlay Memo

Management overlays are described as targeted and modest, capped at 5.0 bps by segment-scenario combina
```

[BANK INPUT] docs/scenario_assumptions.md

Documented scenario narrative that Codex challenged against quantitative reserve behavior and the packaged scenario definitions.

```
# Scenario Assumptions

Baseline, adverse, and severe scenarios were prepared by Finance. Severe is described as a uniformly ha

The review should therefore expect total reserve and segment reserve ordering to be baseline, then adve
```

[CODEX OUTPUT] outputs/support/documentation_crosscheck.md

Codex-generated documentation challenge summarizing the forecast, reversion, overlay, and segment-level inconsistencies identified during the review.

```
# Documentation Cross-Check

## Methodology versus implementation
The written methodology and model overview describe a different forecast and reversion treatment than th

- Evidence: docs/methodology.md, docs/model_overview.md, model/cecl_engine.py
- Documented forecast quarters: 4
- Implemented forecast quarters: 6
- Documented reversion quarters: 4
- Implemented reversion quarters: 2

## Overlay posture
The overlay memo understates the magnitude of the overlay table used in the actual scenario runs.

- Evidence: docs/overlay_memo.md, data/overlay_schedule.csv, outputs/support/sensitivity_results.csv
- Documented overlay cap: 5.0 bps
- Actual overlay cap in schedule: 9.0 bps

## Scenario reasonableness
Overall reserve monotonicity is intact, but one segment behaves oddly under the severe scenario.

- Evidence: outputs/support/segment_reserve_comparison.csv, docs/scenario_assumptions.md, docs/overlay_m
- Overall reserve ordering: baseline 4876841.47, adverse 7132021.97, severe 8287703.47
- Anomaly segment: residential_mortgage
- Adverse versus severe segment delta: -109988.30
```

[CODEX OUTPUT] outputs/support/review_strategy.md

Codex-generated planning record showing how the review questions and procedures were selected from the discovered evidence.

```
## Review Questions
1. Does the package contain enough evidence to support a full execution-based CECL review rather than a
2. Can the supplied baseline reserve be reproduced from the packaged reserve engine, portfolio data, and
3. Do reserve outputs move directionally and materially across Baseline, Adverse, and Severe scenarios b
4. How sensitive are reserves to forecast horizon, reversion speed, macro severity, and overlay magnitud
5. Do the methodology, scenario, and overlay documents faithfully describe the behavior observed in the
```